# Module-3

① Explain matplotlib API, with suitable example?

Matplotlib API :-

- Matplotlib is a popular python library used creating graphs charts & data visualization It is widely used in data Science

The matplotlib API provides functions & classes

| Line plots | function | purpose |
|---|---|---|
| Bar charts | plt.plot() | Create a line plot |
| histograms | plt.bar() | create a bar chart |
| Scatter plots | plt.show() | create a Scatter plot |
| pie charts | plt.hist() | create a histogram. |
| Bar plots | plt.pie() | create a piechart. |

Basic structure of Matplotlib :-

1. Import matplotlib
2. prepare the data
3. Select the type of plot
4. Add filters and labels
5. Customize the plot if required

Advantages :-

1. Easy to use
2. Many plot types
3. Highly customization
4. works with numpy & pandas.

② Different line plot, Barplot, histograms & Scatter plot

Different types in data plot are used for different purpose in data visualization,

- Line plots
- Bar plots
- Histograms plot
- Scatter plot

1. Line plot :-

A plot represents data points connected by lines mainly used to show trends or changes an ordered sequence

```
import matplotlib.pyplot as plt
Months = ["Jan", "Feb", "Mar", "Apr"]
plt.plot [Months, sales]
plt.show()
```

2. Bar plot :-

A Bar plot represents data using rectangular bars. The height or length of each bar represents the value of a category

3. Histogram :-

A histogram is used to represent the distribution of numerical data. It divides data into intervals called bins & shows many observations fall into each interval.

4. Scatter plot :-

A scatter plot displays individual data points using dots. It is mainly used to examine the relationship or correlation b/n numerical variables.

(3) Explain Seaborn facet grid & categorical plots Seaborn

Seaborn is a python data visualization libbuit on top of Matplotlib. It provides a high-level interface for creating attractive & information.

→ It is commonly imported as

```
import seaborn as sns
```

Two imp concepts in Seaborn are

1. face grid
2. Categorical plots.

## Seaborn facegrid:-

facegrid is a Seaborn feature used to create multiple plots based on different subset of a dataset.

### Syntax:

```
Sns.facegrid(data, col = "column")
```

Here data = dataset

col = variable used to create columns.

### Ex:

Suppose we have a dataset containing

Student
Gender
Marks
Department.

Important facegrid parameter

| parameter | purpose |
|---|---|
| Data | Dataset |
| Row | creates separate rows based on a variable |
| column | creates separate columns |
| Hue | uses different colours for categories. |